

Dynamic Geometry-Aware Depth Hypotheses and Edge-Guided Depth Refinement for Aerial Multi-View Stereo

Lina Wang, Jiangfeng She[✉], Zhao Wang, Wei Zhou, and Ben Li

Abstract—Most of the learning-based methods for aerial multi-view stereo (MVS) do not take into account the large depth variations and only rely on coarse stage depth information and uniform sampling to obtain the depth hypotheses, resulting in inaccurate depth estimation. In addition, aerial images cover a large area, and the overlap between multiple views is low, resulting in ambiguous matching in weak texture areas. To address these issues, the DGE-MVSNet is proposed, a novel coarse-to-fine network including a dynamic geometry-aware depth hypotheses (DGDHs) module and edge-guided depth refinement (EGDR) module. In detail, to generate reliable depth hypotheses, the DGDH module creates a neighborhood set for the pixel in the reference view, within which any point should be ensured to consistently lie on the same surface. Relying on this neighborhood set, the initial normal depth hypotheses can be estimated from neighboring points with the same normal. To further ensure accurate depth estimation, the EGDR module utilizes edge feature volumes to guide cost volume aggregation. The model uses cosine similarity to calculate view weights and generate invalidation masks to effectively filter out mismatched regions between source and reference views. During depth refinement, the initial depth is fused with the edge cost volume and combined with the attention block to improve depth estimation accuracy. Extensive experiments on the WHU-MVS and LuoJia-MVS datasets demonstrate that our model significantly outperforms the state-of-the-art methods. Compared to RED-Net, our model reduces the mean absolute error (MAE) by over 24.0%, and by more than 16.0% compared to Cas-MVSNet. To further validate the generalization capability of DGE-MVSNet, the evaluation is conducted on the WHU-OMVS dataset, specifically designed for city 3-D scene reconstruction. Results show that our method achieves a substantial improvement over the state-of-the-art AggrMVS, improving MAE by 29.2%.

Index Terms—Aerial images, deep learning, edge cost volume, geometry-aware, multi-view stereo (MVS).

Received 11 December 2024; revised 19 April 2025 and 9 June 2025; accepted 5 July 2025. Date of publication 8 July 2025; date of current version 22 July 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 42471458 and in part by Nanjing University-China Mobile Communications Group Company Ltd., Joint Institute. (Corresponding authors: Jiangfeng She; Zhao Wang.)

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Digital Object Identifier 10.1109/TGRS.2025.3586962

I. INTRODUCTION

LARGE-SCALE, high-precision 3-D reconstruction of the Earth's surface not only provides realistic virtual environments for virtual reality (VR) and augmented reality (AR) applications but also plays a critical role in practical applications such as urban management, environmental monitoring, and disaster assessment [1], [2]. Although oblique photogrammetry and traditional multi-view stereo (MVS) are the primary algorithms for high-precision 3-D urban reconstruction, they face challenges in low-texture or reflective areas, often requiring substantial manual intervention [3], [4].

In recent years, with the rise of deep learning, researchers have explored learning-based methods for 3-D reconstruction, particularly in tasks like depth estimation and MVS [5], [6], [7], [8]. Inspired by the success of deep convolutional neural networks (CNNs), Yao et al. [9] introduced the first end-to-end deep learning architecture for depth map inference, opening a new field in 3-D reconstruction by leveraging deep models for depth prediction. Learning-based MVS methods [10], [11], [12], [13], [14], [15] have shown significant advantages over traditional methods on multiple public MVS datasets, not only improving accuracy and completeness, but also several times faster. However, most of these methods are optimized for close-range scenes and lack benchmarks for multi-view aerial images, which limits the application and development of deep learning methods in large-scale urban 3-D reconstruction. Some methods have attempted to apply learning-based MVS methods to aerial images and constructed large-scale aerial image benchmark datasets, such as RED-Net [16], MS-REDNet [17], and HDC-MVSNet [18]. Although the above methods have made some progress, they are mainly focused on improving the general MVS framework and have not fully considered the unique challenges of aerial images. Specifically, aerial images often exhibit significant depth range variations (the depth difference between buildings and the ground can reach tens of meters, as shown in Fig. 1(a), which imposes higher requirements on the depth estimation capabilities of existing methods. However, most methods still use a coarse-to-fine depth sampling strategy and the uniform sampling method, which makes it difficult to effectively deal with large-scale depth changes, resulting in insufficient depth estimation accuracy. In addition, aerial images often cover large areas and suffer from severe occlusions, leading to limited overlap between the reference view and the source views [as shown in Fig. 1(a) and (c)]. This further results in sparse matching

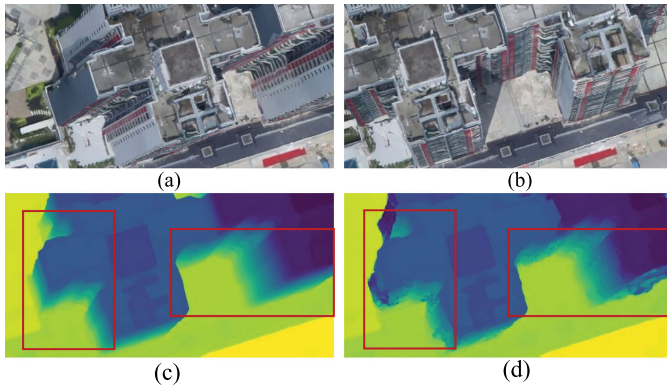


Fig. 1. Aerial images have low multi-view overlap and large depth range variations, which leads to blurred depth estimation in weak texture areas. (a) Reference Image. (b) Source Image. (c) GT. (d) RED-Net.

features and leads to blurred depth estimation compared to the GT, particularly in weakly textured regions such as building edges and occluded areas [as shown in Fig. 1(c) and (d)]. In recent years, several studies have explored effective strategies to tackle the aforementioned challenges. SDL-MVS [19] proposes a deformable learning paradigm across both the view and depth dimensions, which effectively models occlusion and illumination inconsistencies in MVS reconstruction. However, this method primarily focuses on the dynamic adjustment of depth hypotheses and does not fully exploit the geometric information associated with depth inference. EG-MVSNet [20] introduces an edge-aware mechanism to improve depth estimation near building boundaries. Nonetheless, it remains centered on enhancing structural contours using edge information, while overlooking the deeper role that edge features can play in optimizing depth accuracy. Moreover, its high-computational cost limits its applicability in real-world scenarios.

To tackle these issues, a coarse-to-fine structure DGE-MVSNet is proposed, including a dynamic geometry-aware depth hypotheses (DGDHs) module and an edge-guided depth refinement (EGDR) module. The DGDH module uses adaptive propagation to generate a set of neighborhoods for the points in the reference image, ensuring that each point within the neighborhood lies on the same surface. Based on this neighborhood set, the initial normal depth hypotheses are estimated using neighboring points with the same normal. The uncertainty awareness of each neighboring point is then weighted and averaged to optimize the normal depth hypotheses. The EGDR module groups features along the channel dimension when aggregating the cost volume. The view weight and invalid mask are calculated based on the cosine similarity to filter the invalid areas where the source view cannot match the reference view. The initial depth is fused with the edge cost volume and combined with the attention mechanism to improve the depth estimation accuracy in the coarse stage. Extensive experiments on multiple benchmark datasets [18], [21], [22] show that our method exhibits superior depth visualization on the aerial MVS dataset compared to all listed methods, especially in weak texture areas such as edges or bottoms.

The main contributions are summarized as follows.

1) The DGDH module is based on geometric consistency and uses uncertainty awareness to dynamically optimize the normal depth hypotheses, significantly improving the accuracy of the depth hypotheses.

2) The EGDR module computes view weights and generates invalidation masks during cost volume aggregation, effectively filtering out areas that do not match between the source view and the reference view. During depth refinement, the module fuses the coarse depth with the edge cost volume and combines the attention block to improve depth estimation accuracy.

3) Extensive experiments on public benchmarks, such as WHU-MVS and LuoJia-MVS datasets, show that our DGE-MVSNet method achieves superior performance compared to SOTA methods.

II. RELATED WORK

A. Learning-Based MVS for Close-Range Reconstruction

Recently, many studies have applied learning-based MVS methods, which have achieved significant performance improvements over the traditional methods on close-range reconstruction. Yao et al. [9] proposed a depth map-based method, MVSNet, which is an end-to-end trainable architecture with three steps: feature extraction, depth hypotheses, and cost volume. Early methods [21], [23], [24] used single-scale predicted depth maps, which required a large number of depth hypothesis planes to achieve better performance. However, these methods were constrained by memory bottlenecks, making it difficult to construct high-precision cost volumes from high-resolution images. Recently, a novel coarse-to-fine strategy has been widely adopted in MVS tasks to reduce memory consumption. Cas-MVSNet [10] and CVP-MVSNet [12] proposed cascaded cost volumes based on a feature pyramid network, estimating depth maps in a coarse-to-fine manner. Before computing the cost volume, accurately estimating the depth hypotheses range is crucial. UCS-Net [25] employs variance-based uncertainty estimation to adaptively construct adaptive thin volumes (ATVs), achieving high completeness and accuracy in scene reconstruction through a coarse-to-fine architecture. PatchmatchNet [11] employs attention mechanisms to calculate view weights, thereby suppressing the adverse effects of occluded pixels in the reference view. CostFormer [15] uses a Transformer to achieve efficient cost aggregation and solves the problem of incorrect MVS matching. The success of CNN in close-range MVS tasks has sparked a wave of research and exploration in the field of large-scale aerial MVS. Motivated by these methods, aerial MVS approaches have seen continuous development.

B. Large-Scale Aerial MVS

Although learning-based methods have achieved remarkable success in close-range reconstruction tasks [26], [27], [28], [29], applying MVS technology to large-scale aerial images still faces many challenges, mainly due to the wide range of depth variations, diverse scene types, and significant viewpoint differences in aerial scenes. In recent years, some studies have made important progress in achieving efficient multi-view depth estimation of aerial images. RED-Net [16]

is the first MVS network designed specifically for aerial image depth estimation. Its performance is better than that of traditional aerial MVS methods, but the improvement is limited, and it is difficult to cope with complex aerial scenes. MS-REDNet [17] combines MVS and semantic segmentation to achieve high-precision reconstruction of buildings, but still lacks a clear and effective depth estimation strategy. HDC-MVSNet [18] enhances context information modeling through a hierarchical deformable cascade structure to achieve full-resolution depth map generation, but still lacks effective perception of depth information in aerial images. Ada-MVS [22] proposed a 3-D scene reconstruction framework, Deep3D, which generates textured 3-D mesh models from multi-view oblique aerial images based on MVS matching. However, this method is limited in large-scale aerial applications by significant viewpoint changes and insufficient overlap of adjacent views, resulting in blurred matching of edge regions, which increases the difficulty of recovering fine structures and object boundaries. To enhance the matching accuracy around building edges, EG-MVSNet [20] incorporates edge information by jointly predicting depth and edge maps, which effectively improves reconstruction performance in edge regions. However, this method primarily focuses on enhancing edge features, without fully exploiting the role of edge information in optimizing the depth map. Moreover, its high computational cost limits its applicability, especially in scenarios involving large depth ranges, such as aerial imagery. Additionally, dc-SatMVS [30] proposes a contrastive learning-based distributed satellite MVS network tailored for surface depth estimation in satellite images, but its performance still lags behind methods specifically designed for aerial data. Several studies [30], [31] have attempted to integrate general MVS modules with popular pretrained models such as Depth-anything. Nevertheless, these approaches still fall short in fully addressing the inherent complexities and challenges of aerial image reconstruction.

C. Depth Hypotheses

The core of MVSNet is the matching process between reference pixels and source pixels, and depth hypotheses play an important role in this process. Cas-MVSNet [10] proposed to use cascaded cost volumes to pass depth hypotheses to the next stage. However, this method still relies on uniform sampling and cannot deal with the weak texture area information of the depth hypothesis. PatchmatchNet [11] proposes an adaptive propagation algorithm to collect depth hypotheses from the same physical surface, which helps the network converge faster and provide more accurate depth maps. UCS-Net [25] variance-based uncertainty estimation to adaptively construct ATVs, gradually subdividing the vast scene space to achieve scene reconstruction with high completeness and accuracy in a coarse-to-fine manner. GBi-MVSNet [32] proposes a binary search network to avoid error accumulation of erroneous network predictions and improve efficiency. ARAI-MVSNet [33] proposes adaptive depth range prediction and depth interval adjustment modules to achieve more reliable depth estimation. However, these methods often ignore the geometric information contained in the depth hypothesis and

usually rely only on the depth estimation of the coarse stage to determine the depth search range of the next stage. In large-scale aerial images, due to challenges such as wide viewing angles and drastic depth changes, it is difficult to effectively capture the complex geometric relationships within the scene using only coarse depth information. This limitation leads to reduced accuracy of subsequent depth estimation and increases the risk of error accumulation. To the best of our knowledge, there is currently no depth range estimation method specifically for aerial MVS reconstruction. Therefore, this article proposes a new depth range estimation strategy, which uses adaptive propagation sampling instead of uniform sampling based on the cascade structure of Cas-MVSNet, and integrates geometric information and uncertainty perception to improve the accuracy and robustness of depth estimation for large-scale aerial images.

D. Edge Aware Refinement

Edge features are important information in computer vision and image processing [34], [35], [36], and play a key role in tasks such as image classification [37], [38], object detection [39], [40], [41], and segmentation [42]. In recent years, edge information has been gradually introduced into the field of 3-D reconstruction and stereo vision, and applied to MVS tasks to improve the accuracy and stability of depth estimation. DispNet [43] reduces the occlusion fading problem in self-supervised monocular depth estimation through edge-guided post-processing, and proposes a lightweight depth estimation network based on ASPP, which has faster reasoning speed and better performance. DEF-MVSNet [44] adopts deep edge flow to dynamically infer edge coordinates guided by reference images, thereby improving the integrity of 3-D reconstruction. EPNet [45] designs an efficient edge-preserving MVS network for accurate depth estimation, but its edge information is only used as an auxiliary feature and has not yet played a role in the core calculation of the MVS process. EG-MVSNet [20] proposes an end-to-end edge-aware deep reasoning network dedicated to MVS tasks of large-scale aerial buildings. However, the main goal of this method is to optimize the edge detection of buildings rather than directly improve the rough depth estimation, and there is a high memory consumption problem. Inspired by these methods, this study incorporates edge information into the MVS processing pipeline and combines the attention mechanism to fuse edge feature volumes with coarse depth to optimize the depth refinement process and improve the accuracy and robustness of 3-D reconstruction.

III. METHODOLOGY

Our approach extends RED-Net [19] into a cascade network DGE-MVSNet for large-scale aerial image reconstruction, incorporating geometry-aware and edge features to enhance depth hypotheses and improve depth map accuracy in the coarse stage. The network architecture is shown in Fig. 2, which mainly includes multiscale feature extraction (MFE), DGDHs, and EGDR modules, among which DGDH and EGDR are our main contributions.

MFE: The method takes the reference image I_0 , the source images $\{I_i\}_{i=1}^N$ and its corresponding camera extrinsics $\{T_i\}_{i=0}^{N+1}$ and $\{K_i\}_{i=0}^{N+1}$ as input. The image features $\{F_i\}_{i=0}^{N+1}$ and edge features $\{F_{i,e}\}_{i=0}^{N+1}$ are obtained through the MFE module.

DGDH: The module is used to refine the depth hypothesis range and is divided into three stages: the first stage uses uniform sampling to obtain the initial depth hypothesis; the second and third stages use the proposed DGDH module to refine the depth hypothesis more accurately.

EGDR: Based on the camera internal and external parameters and depth hypothesis, the image feature volume $\{V_i\}_{i=0}^{N+1}$ and edge feature volume $\{V_{i,e}\}_{i=0}^{N+1}$ are first calculated using differentiable homography. To incorporate edge information into cost volume aggregation, this method proposes an edge-guided depth refinement module: first, the feature volumes are grouped by channels and the cost volumes are aggregated using cosine similarity; then, the aggregated cost volumes are regularized and depth regressed to generate an initial depth map; finally, the edge feature volume is fused with the initial depth map, and an attention module is integrated to optimize the depth refinement.

A. MFE Module

Most of the networks used for large-scale aerial image reconstruction in the past are based on RED-Net [22]. To improve the network efficiency and accuracy, the MFE module is designed, including a cascaded network [46] for image features $F \in \mathbb{R}^{C \times H \times W}$ and an edge detection network for edge features $F_e \in \mathbb{R}^{C \times H \times W}$, as shown in Fig. 3. The image feature extraction network includes an encoder and a decoder. The encoder consists of three convolution units, each of which consists of a series of convolution layers. Except for the last layer, each convolution layer is followed by a batch normalization layer and a ReLU activation layer. The decoder consists of three deconvolution units. Except for the deconvolution layer, the components of the deconvolution unit are the same as those of the convolution unit. In order to compensate for the information lost in the encoder, skip connections are used to fuse feature maps of the same size between the encoder and the decoder. The edge feature extraction network mainly includes an edge detection module as well as a decoder and an encoder. Our method tries to reduce memory consumption as much as possible while ensuring accuracy (for example, architectures such as EG-MVSNet [20] have high memory usage). Therefore, a Canny algorithm [47] based on median filtering is selected. Compared with the traditional Canny algorithm, this method can not only extract edge features more efficiently, but also maintain a low model complexity. Specifically, the image is first smoothed using median filtering to reduce noise interference; then, the Sobel operator is used to calculate the intensity and direction of pixel gradients; then, nonmaximum suppression is used to remove redundant edge responses; finally, the double threshold method is used to filter edge information to improve the robustness of detection.

B. Dynamic Geometry-Aware Depth Hypotheses

Large-scale aerial images exhibit significant depth range variation, which poses challenges for depth hypotheses

methods. Most traditional approaches depend on uniform sampling and coarse-stage depth estimation, leading to unreliable depth hypotheses. To address this, our work introduces the DGDH module, aimed at generating geometry-aware depth hypotheses related to depth. This approach accurately predicts depth hypotheses for occluded and low-texture areas, enhancing overall depth hypotheses accuracy.

Initially, we create a set of neighborhoods for the pixel in reference view. These neighborhoods consist of points that consistently lie on the same physical surface, enabling more precise depth prediction across varied and complex surfaces [15]. Specifically, for each point p , an initial set of neighborhood coordinates $p_{j=1}^{k_p}$ is defined. The reference feature map is then processed using dilated convolution to derive offsets for each neighborhood point $\Delta o(p_j)_{j=1}^{k_p}$. These offsets are added to the initial neighborhood coordinates to generate adaptive neighborhood coordinates $k_j = \Delta o + p_{j=1}^{k_p}$. Using this adaptive neighborhood, the initial normal depth hypotheses are estimated based on neighboring pixels with a shared surface normal. To compute the normal of this neighborhood set, a plane is fit through these points using the least squares method

$$n_j = \frac{(K^T K)^{-1} K^T \mathbf{1}}{\| (K^T K)^{-1} K^T \mathbf{1} \|_2} \quad (1)$$

where $K = \begin{bmatrix} k_1^T \\ k_2^T \\ \vdots \\ k_p^T \end{bmatrix}$ represents the matrix formed by the set of

neighborhood points k_j , $\mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}$ denotes a matrix used to calculate the plane in surface fitting.

Furthermore, we project the neighborhood point onto the tangent plane corresponding to the normal to obtain the corresponding projection point k'_j

$$k'_j = k_j + n_j \frac{n_j \cdot (k_j - k)}{\| n_j \|_2}. \quad (2)$$

The normal depth range is determined based on the projection point and the intrinsic parameters. Specifically, the intrinsic matrix is multiplied by the projection matrix, and relevant values are extracted to yield the projected depth information. The depth information is then expanded to match the dimensions of the corresponding neighborhood points, resulting in the normal depth range D_n .

Uncertainty awareness is a method to reduce the depth refinement range [25]. This method can accurately perceive pixels with reliable probability values and obtain a better depth refinement range. The standard deviation of the probability distribution at pixel x and stage m is calculated as follows:

$$\sigma_m(x) = \sqrt{\sum_{j=1}^{d_m} P_{m,j}(x) \cdot (L_{m,j}(x) - \hat{L}_m(x))^2} \quad (3)$$

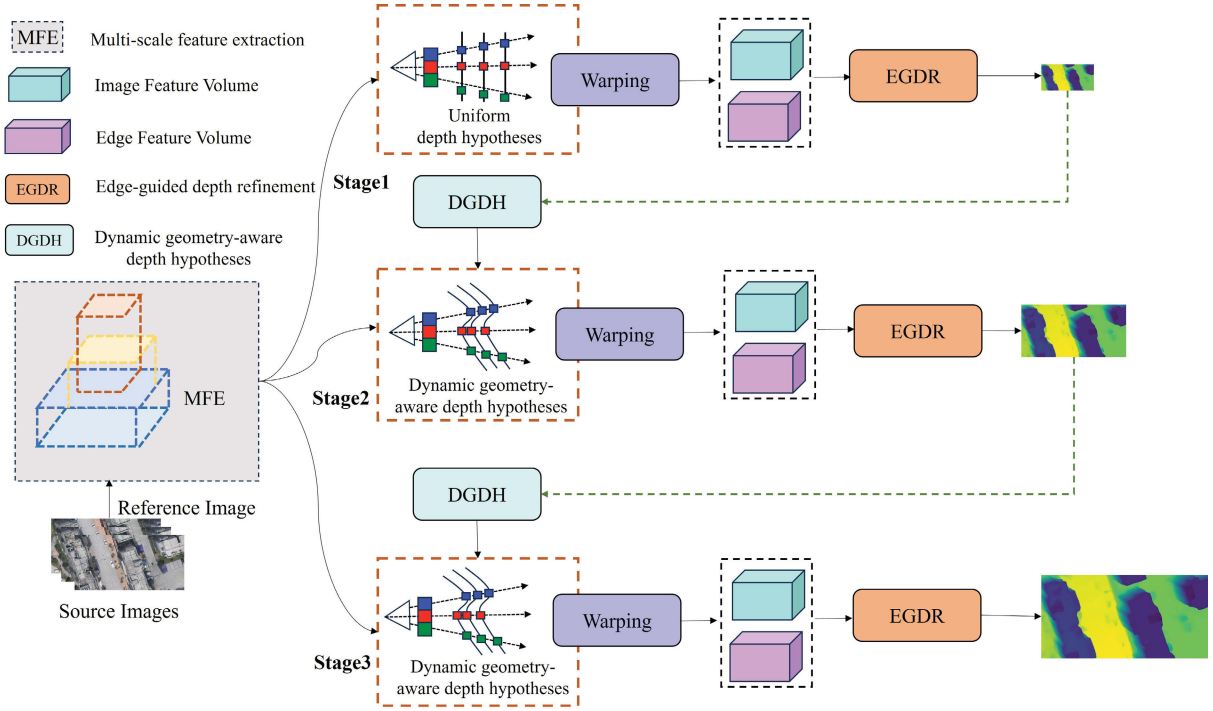


Fig. 2. Architecture of DGE-MVSNet. Our network consists of three parts, including MFE, DGDH, and EGDR, which obtain multiscale features, depth hypothesis range, and refined depth, respectively. The DGDH and EGDR modules are our proposed method, and the details are shown in Sections III-B and III-C.

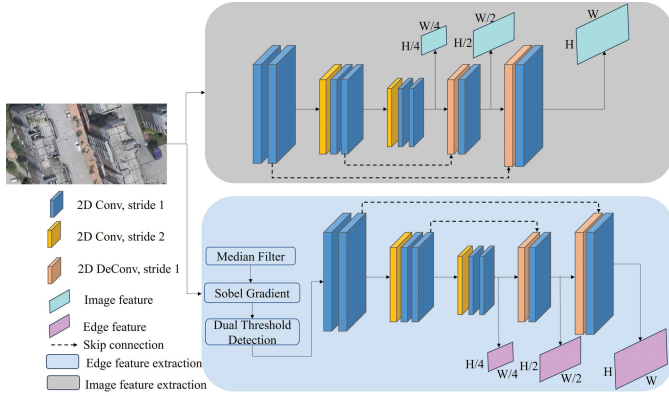


Fig. 3. Illustration of the MFE module is mainly composed of an image feature extraction network and an edge feature extraction network.

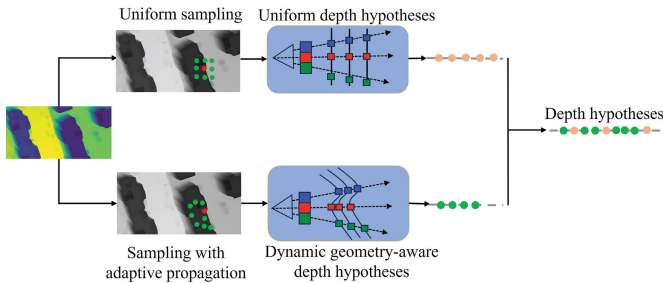


Fig. 4. Structure of the DGDH module. This module mainly consists of two parts: uniform sampling and dynamic geometry-aware sampling. Uniform sampling generates uniform depth hypotheses through a fixed strategy, while dynamic geometry-aware sampling generates normal-guided depth hypotheses based on adaptive propagation.

where d_m represents the number of planes for stage m , $L_{m,j}(x)$ represents the depth hypotheses of the j th plane at pixel x in

stage m , $P_{m,j}(x)$ represents the probability that the depth at pixel x is $L_{m,j}(x)$, and $\hat{L}_m(x)$ represents the depth prediction at pixel x .

However, the probability volume can be inaccurate in edge or occluded regions, where incorrect normal sampling intervals may be generated. To address this issue, we employ dynamic propagation to update the probability distribution of the normal depth hypotheses. This is achieved by calculating the standard deviation of the center point and neighboring points and applying weighted adjustments

$$\hat{\sigma}_m(x) = \lambda_1 \hat{\sigma}_0(x) + \lambda_2 \sum_{j=1}^{k_p} \hat{\sigma}_{m,j}(x) \quad (4)$$

where $\hat{\sigma}_0(x)$ is the standard deviation of the center point, $\hat{\sigma}_{m,j}(x)$ is the standard deviation of the neighborhood points, k_p is the number of neighbor points, and λ_1 and λ_2 are variance weights.

Based on uncertainty awareness, we obtain the depth sampling range $\hat{C}_m = [\hat{L}_m(x) - \lambda \hat{\sigma}_m(x), \hat{L}_m(x) + \lambda \hat{\sigma}_m(x)]$, and combine the sampling interval to obtain the normal depth hypotheses $\hat{D}_{n,m+1}$ of the next stage. Finally, the union of the dynamic normal depth hypotheses and the uniform sampling depth hypotheses is obtained as the refined depth hypotheses

$$D_{m+1} = \hat{D}_{n,m+1} \cup D_{u,m+1}. \quad (5)$$

C. Edge-Guided Depth Refinement

Due to the low overlap of multi-view features, existing learning-based MVS methods have difficulty in achieving robust feature matching, so it is crucial to rationally utilize contour information to enhance features and improve the our method proposes an edge-guided depth refinement module,

which uses edge features to guide depth estimation and combines attention to improve the accuracy of depth estimation in the coarse stage. The feature cost volume is further constructed based on the image features $F \in \mathbb{R}^{C \times H \times W}$ and edge features $F_e \in \mathbb{R}^{C \times H \times W}$, and all source views are warped into the reference view through homography to generate N image cost volumes $V \in \mathbb{R}^{C \times D \times H \times W}$ and edge cost volumes $V_e \in \mathbb{R}^{C \times D \times H \times W}$. Suppose the pixel coordinate of the reference view is x_0 and the depth is d_0 . The projected coordinates of the reference view at the i th source are calculated as follows:

$$x_j = K_j R_j ((R_0^T K_0^{-1} x_0 d_0) + (R_j^T T_j - R_0^T T_0^l)) \quad (6)$$

where K_0, K_j, T_0, T_j , and R_j are the camera intrinsic matrix, rotation matrix, and translation vector of the reference view and the i th source view, respectively; x_j is the homogeneous coordinate after x_0 mapping.

The EGDR module proposes a new cost volume aggregation method based on cosine similarity. First, the feature volume is divided into G groups according to the channel, and the grouped feature volume $V \in \mathbb{R}^{G \times (C/G) \times D \times H \times W}$ is obtained. Each group of features is regarded as a vector, and the similarity of the feature vector can be calculated using the inner product. V_0 and F_i are the feature volumes of the reference view and the i th source view, respectively. The inner product of the reference view and the i th source view in the g group is calculated as follows:

$$S_i = \langle F_{0,g}, F_{i,g} \rangle \quad (7)$$

where $\langle \cdot, \cdot \rangle$ represents the inner product.

The view weight is calculated based on the cosine similarity as

$$W_{i,g} = \frac{S_{i,g}}{\|F_{i,g}\| \|F_{0,g}\|}. \quad (8)$$

Similar to other methods of calculating view weights, it is input into continuous $1 \times 1 \times 1$ 3-D convolution for filtering and fusion, and finally the value is mapped between $[0,1]$ through the Sigmoid function to obtain the weight $W_{i,g} \in \mathbb{R}^{D \times H \times W}$.

There are regions in the source view that cannot be matched with the reference view, contributing negatively to feature matching. These regions exhibit low feature similarity, leading to uniform computed view weights. We smooth the view weights by 3×3 average pooling to remove noise and enhance feature stability. Then, we calculate the difference between the smoothed weights and the original weights. If the difference is lower than a set threshold, the corresponding pixel is retained; otherwise, it is set to 0

$$\text{Mask} = |\text{AveragePool}(W_{i,g}) - W_{i,g}| < \delta \quad (9)$$

where AveragePool represents average pooling, δ represents the threshold. The calculated mask is multiplied by the weight to get the valid weight

$$W_{\text{mask}} = \text{Mask} \otimes W_{i,g}. \quad (10)$$

Calculate the weighted average of each perspective to obtain the aggregate cost volume C

$$C = \frac{\sum_{i=1}^N W_{\text{mask}} \cdot S_i}{\sum_{i=1}^N W_{\text{mask}}}. \quad (11)$$

According to the cost volume aggregation method, we obtain the image cost volume $C \in \mathbb{R}^{1 \times D \times H \times W}$ and the edge cost volume $C_e \in \mathbb{R}^{1 \times D \times H \times W}$.

The original cost volume computed from image features may be contaminated by noise, and a smoothness constraint should be incorporated to infer the depth map. We adopt recurrent cost volume regularization [20] to convert the cost volume into a probability volume $P \in \mathbb{R}^{1 \times D \times H \times W}$ and perform depth regression [9] to generate the initial depth map D_{init}

$$D_{\text{init}} = \sum_{d=d_{\min}}^{d=d_{\max}} d \times P(d) \quad (12)$$

where $[d_{\min}, d_{\max}]$ represent the depth hypotheses.

To obtain the global information of the attention edge cost volume, the edge cost volume C_e and the initial depth map D_{init} are connected and sent to the attention module. The working process of the attention module is shown in Fig. 5. First, a convolutional network is used to reduce the channels of the fused cost volume and encode it as the input of the attention. Then, the output of the convolutional network is sent to the attention encoder to obtain the corresponding features, and finally, multiplied by the initial edge cost volume to obtain the attention edge cost volume. To further refine the depth map, the attention edge cost volume is regularized, and the depth regression is performed to obtain the attention depth map, which is finally added to the initial depth map to obtain the refined depth map D_{refine} .

D. Loss Function

Our model adopts a coarse-to-fine feature pyramid framework, so we use a loss function similar to Cas-MVSNet [10], which is defined as the smooth loss L_1 between the estimated depth map D_{refine} and the true depth map D_{gt} at each stage

$$L = \sum_{k=1}^3 \lambda^k \|D_{\text{refine}}^k - D_{\text{gt}}^k\| \quad (13)$$

where λ represents the loss weight of each stage.

IV. EXPERIMENTS AND RESULTS

In this section, we train our model on the WHU-MVS Training set, and evaluate our model on the WHU-MVS Testing set [16], LuoJia-MVS dataset [18], and WHU-OMVS dataset [22]. First, we introduce the training data and testing datasets. Then, we introduce the evaluation metrics and training parameter settings, and we compare the evaluation accuracy and efficiency of our model with other similar methods. To further verify the effectiveness of our proposed module, we conduct ablation experiments.

A. Aerial MVS Datasets

1) *WHU-MVS Dataset*: WHU-MVS dataset [16] is a large-scale aerial image dataset created specifically for large aerial MVS tasks, generated by high-precision 3-D digital models. It is a set of artificial aerial images taken from an

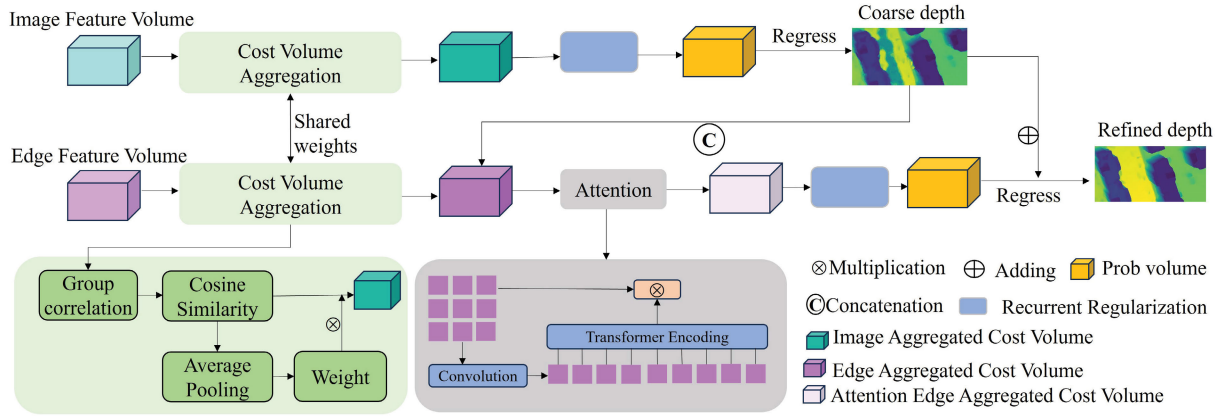


Fig. 5. EGDR module mainly includes cost volume aggregation and depth regression. Cost volume aggregation calculates view weights based on group correlation and cosine similarity, and depth regression relies on edge cost volume to refine the coarse depth.

area of about 6.7×2.2 square kilometers, covering different types of scenes, including dense high-rise buildings, sparse factories, forest-covered mountains, bare land, and rivers. The deep learning subdataset consists of 4320 pairs of five-view images with a spatial resolution of 10 cm, each image is 768×384 pixels, and the ratio of training set to test set is roughly 3:1.

2) *LuoJia-MVS Dataset*: LuoJia-MVS dataset [18] is also an aerial image dataset dedicated to multi-view 3-D reconstruction tasks. It is composed of 1430 pairs of two-view aerial images constructed by a series of software tools such as Photoscan, Smart3D, and generated by 3-D DSM and OpenSceneGraph binary (OSGB) format meshes. The deep learning training set and test set are 4320 and 1360 multi-view units, respectively. Each view consists of an RGB image with a size of 768×384 pixels and a resolution of 10 cm and its pixel-by-pixel depth map. The dataset covers cultivated land, forest land, towns, villages, industrial and mining areas, residential land, unused land, etc., further providing a new benchmark for depth estimation of MVS aerial images.

3) *WHU-OMVS Dataset*: The WHU-OMVS dataset [22] is a large-scale synthetic multi-view oblique aerial image dataset, which is mainly used for city-level 3-D scene reconstruction. It uses the same basic scene rendering as the WHU-MVS dataset, expanding the original bottom-view dataset into an oblique five-view stereo dataset. The aerial images are taken by an oblique five-view camera system mounted on an uncrewed aerial vehicle (UAV) at an altitude of about 550 m, with image overlap of 80% in the flight direction and 60% in the flight direction, and a ground resolution of about 10 cm. The 3-D scene model area is about 6.7×2.2 square kilometers, covering a typical urban area with rich details. The training set and validation set consist of 768×384 pixel rendered images in PNG format, the corresponding depth map is in float EXR format, and the imaging parameters of each image are in TXT format.

B. Metrics and Implementations

1) *Metrics*: Our method uses several common metrics to evaluate the quality of the depth map as follows.

a) *Mean absolute error*: the average of the $L1$ distances between the estimated depth and the true depth, and only the distances within 100 depth intervals are calculated to exclude extreme outliers.

b) $\sigma < 0.6$ m: the percentage of pixels with an $L1$ error less than the 0.6 m threshold.

c) $\sigma < 3$ -interval: the percentage of pixels with an $L1$ error less than three depth intervals. Among these metrics, a lower mean absolute error (MAE) value indicates higher accuracy, while higher values of $\sigma < 0.6$ m and $\sigma < 3$ -interval reflect better depth estimation performance.

2) *Training Implementations Details*: We have implemented our proposed DGE-MVSNet using PyTorch and trained it on the WHU-MVS dataset. We set the number of input images to $N = 5$ with a resolution of 768×384 . The selection of reference images and source images is the same as that of RED-Net. Similar to Cas-MVSNet [10] with three stages of 1/4, 1/2 and full image resolutions, the initial depth hypotheses range $[d_{\min}, d_{\max}]$ is obtained from the SfM sparse scene, the number of depth hypotheses is 48, 32, and 8 for each stage, the weight of each stage loss λ^k is set to 0.5, 1.0, and 2.0, and the corresponding depth interval is set to 4, 2, and 1, respectively. The training is performed on 1 Nvidia RTX 4090 GPU for 16 epochs with a batch size of 1.

3) *Testing Implementations Details*: Some basic settings remain the same as training. We input the image size as 768×384 and set the number of views to $N = 5$ and $N = 3$, respectively. We adopt the aforementioned evaluation metrics to evaluate the quality of the predicted depth maps.

C. Benchmarking on WHU-MVS Dataset

We evaluate our model on the WHU-MVS testing dataset and compare it with other learning-based methods. As shown in Table I, by incorporating a dynamic geometry-aware depth assumption and edge cost, our model achieves the best performance on both the five-view and three-view WHU-MVS datasets. Specifically, compared to close-range MVS methods, our approach demonstrates notable improvements. For example, in the five-view scenario, the MAE is reduced from 0.160 with PatchmatchNet [11] to 0.079, marking a 50.6%

TABLE I

COMPARISON OF QUANTITATIVE RESULTS OF DIFFERENT MVS METHODS ON THE WHU-MVS DATASET

Number of views	Method	MAE (m)	$\sigma < 3$ -interval (%)	$\sigma < 0.6m$ (%)
Three-view	PatchmatchNet [11]	0.173	94.8	96.5
	MVSNet [9]	0.190	94.3	95.0
	R-MVSNet [21]	0.183	93.5	95.3
	RED-Net [16]	0.112	97.9	98.1
	Cas-MVSNet [10]	0.111	97.6	97.7
	HDC-MVSNet [18]	0.101	97.8	97.9
	SDL-MVS [19]	0.095	98.4	98.0
	EG-MVSNet [20]	0.097	98.0	98.2
	AggrMVS [48]	0.155	97.1	98.1
	DGE-MVSNet (Ours)	0.095	98.1	98.4
Five-view	PatchmatchNet [11]	0.160	95.0	96.9
	MVSNet [9]	0.160	95.5	95.8
	R-MVSNet [21]	0.173	93.8	95.4
	RED-Net [16]	0.104	97.9	98.1
	Cas-MVSNet [10]	0.095	97.8	97.8
	HDC-MVSNet [18]	0.087	98.0	98.1
	SDL-MVS [19]	0.082	98.7	98.9
	EG-MVSNet [20]	0.081	98.7	98.5
	AggrMVS [48]	0.103	98.0	98.6
	DGE-MVSNet (Ours)	0.079	98.9	99.1

improvement. Similarly, the MAE of Cas-MVSNet [11] is improved from 0.095 to 0.079 (an improvement of 16.8%). Moreover, when compared to other aerial image depth estimation methods such as RED-Net [16] and HDC-MVSNet [18], our method decreases the MAE from 0.104 and 0.087 to 0.095, respectively.

In addition, we conduct a qualitative visualization comparison of Cas-MVSNet [11], RED-Net [16], and EG-MVSNet [20] with the proposed DGE-MVSNet, and the results are shown in Fig. 6. From the visualization results, the proposed method has a better depth estimation effect in edge areas and bottom easily occluded areas (red local enlarged area). This is because the proposed method fully integrates edge features to guide the depth refinement process, thereby significantly alleviating the matching blurriness, making the boundary clearer, and the structural integrity stronger. In addition, compared with methods such as EG-MVSNet, the proposed method not only effectively integrates edge information into the depth estimation process but also further proposes DGDH, which has a better reconstruction effect in weak texture occlusion areas. In summary, both quantitative and qualitative experimental results fully verify the effectiveness and advantages of DGE-MVSNet in aerial multi-view scenarios, especially in edge and occlusion.

D. Benchmarking on Other Aerial MVS Datasets

1) *Evaluation on the LuoJia-MVS Dataset:* To demonstrate the generalization capability of our method, we validated our model on the LuoJia-MVS [18] dataset. Some statistics in Table II are based on HDC-MVSNet [18]. For a fair comparison, we configured our evaluation following the specifications outlined in HDC-MVSNet. Our method consistently achieved top scores across all evaluation metrics. Compared to other methods not specifically designed for aerial

TABLE II

QUANTITATIVE COMPARISON OF DIFFERENT MVS METHODS ON THE LUOJIA-MVS DATASET

Number of views	Method	MAE (m)	$\sigma < 3$ -interval (%)	$\sigma < 0.6m$ (%)
Three-view	PatchmatchNet [11]	0.252	87.2	92.7
	MVSNet [9]	0.172	92.4	96.1
	R-MVSNet [21]	0.177	93.5	96.0
	RED-Net [16]	0.109	96.9	98.2
	Cas-MVSNet [10]	0.103	97.1	98.4
	HDC-MVSNet [18]	0.089	97.8	98.7
	SDL-MVS [19]	0.086	98.9	98.9
	EG-MVSNet [20]	0.087	97.9	98.9
	DGE-MVSNet (Ours)	0.095	98.9	99.2
	PatchmatchNet [11]	0.283	84.1	90.4
Five-view	MVSNet [9]	0.270	81.8	91.2
	R-MVSNet [21]	0.259	86.7	92.3
	RED-Net [16]	0.156	90.5	94.9
	Cas-MVSNet [10]	0.141	95.4	97.9
	HDC-MVSNet [18]	0.121	96.6	98.3
	SDL-MVS [19]	0.118	98.6	98.2
	EG-MVSNet [20]	0.115	96.9	98.4
	DGE-MVSNet (Ours)	0.110	98.0	98.8

images [1], [9], [10], [21], our approach leverages dynamic uncertainty-aware geometry to achieve refined depth estimation ranges, which enhances depth map quality. For example, our method improves MAE from 0.252 to 0.095 (62% improvement) for PatchmatchNet [11], and from 0.177 to 0.095 (46% improvement) for R-MVSNet [21] in three-view. In addition, Fig. 7 shows depth map visualizations. Traditional methods like RED-Net [16] struggle with accurate depth range prediction, resulting in blurry matching in weak texture areas, incomplete depth estimation compared to ground truth. Our method effectively addresses these limitations by utilizing dynamic uncertainty-aware geometry for depth hypotheses and edge-guided depth refinement, achieving significantly more precise depth map estimation.

2) *Evaluation on the WHU-OMVS Dataset:* In addition, to further demonstrate the generalization ability of DGE-MVSNet, we also evaluate it on the WHU-OMVS [22] dataset for city-level 3-D scenes. Some statistics in Table III are obtained from AggrMVS [48]. For fair comparison, we retrained our model on the WHU-OMVS [22] dataset with the same parameter settings as AggrMVS [48] (The model was trained for 16 epochs, with the initial learning rate set to 0.001). In addition, AggrMVS [18] has no public code, and only the accuracy comparison is performed without a visual comparison. Table III gives the accuracy comparison of DGE-MVSNet and other methods on the WHU-OMVS [22] dataset. DGE-MVSNet pretrained on the WHU-OMVS dataset shows excellent performance in all evaluation indicators. Impressively, our model outperforms the SOTA AggrMVS [48] by a large margin, with an improvement of 29% on MAE. This significant enhancement can be attributed to the enhancement of building outline information by DGE-MVSNet. Fig. 8 shows the comparison between the depth map generated by DGE-MVSNet and other methods. It can be clearly seen that the depth prediction of the traditional methods Cas-MVSNet [10] and Ada-MVS [22] at the edge

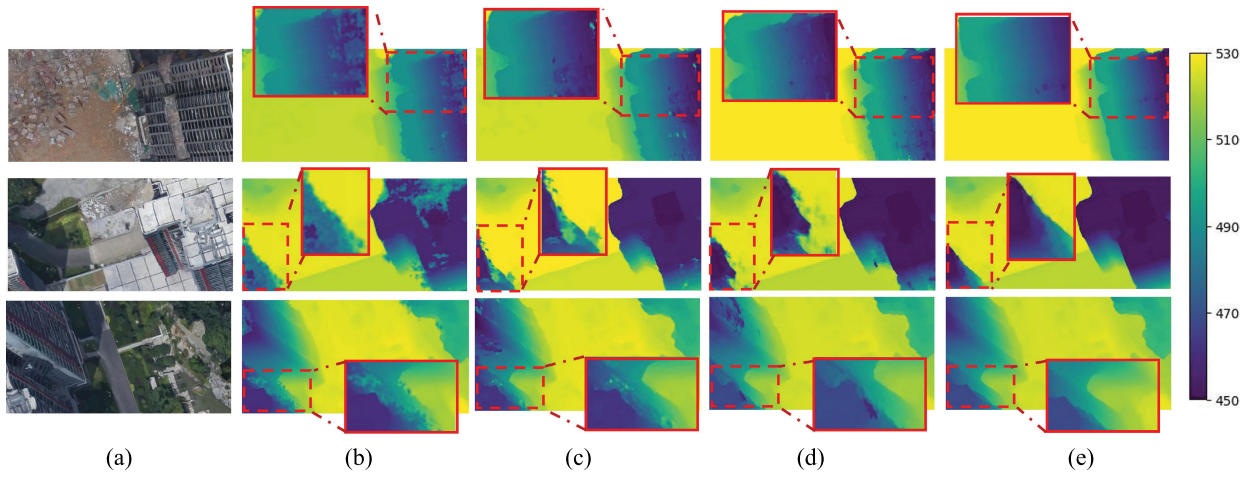


Fig. 6. Qualitative results of mainstream methods on three different regions of the WHU-MVS dataset [16]. Our method obtains more complete and smoother depth maps, especially in the edge region and bottom occluded area, as shown in the local magnified area of the red bounding box. The scale of the color bar represents the depth value. (a) Reference Image. (b) Cas-MVSNet. (c) RED-Net. (d) EG-MVSNet. (e) Ours.

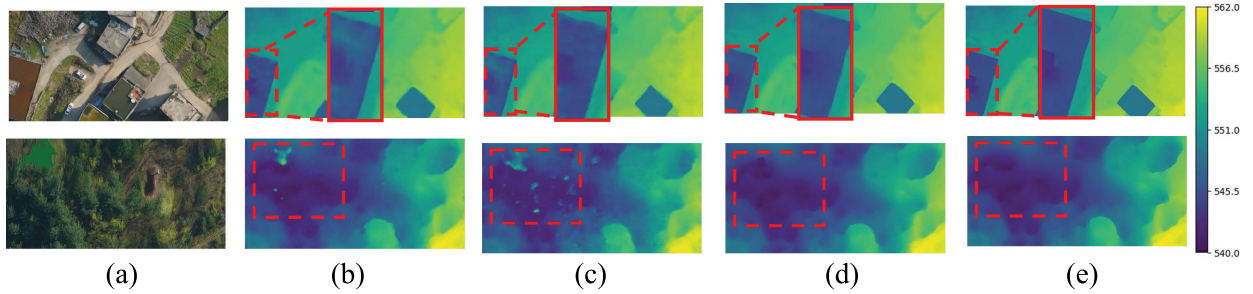


Fig. 7. Qualitative comparison results of different MVS methods on the LuoJia-MVS dataset [18]. In complex scenes (such as forests and villages), our method can still generate more complete and detailed depth maps. (a) Reference Image. (b) Cas-MVSNet. (c) RED-Net. (d) Ours. (e) GT.

of the building and the occluded area is often incomplete or blurred, while DGE-MVSNet significantly improves the accuracy of depth estimation by effectively integrating geometric information and edge features, and generates a depth map with a more complete structure and clearer boundaries. The experimental results show that this method has good generalization ability in different scenarios and can adapt to various types of data, further verifying its robustness and applicability.

E. Ablation Study

To further demonstrate the contributions of our DGE-MVSNet, we present an ablation study on the WHU-MVS test dataset (five-view) in this section.

1) *Benefit of Each Component*: Four experiments were conducted here to illustrate the improvements brought by each component of our method, as shown in Table IV. First, we removed the DGDH and EGDR modules from DGE-MVSNet to create a baseline model. Then, we added each module individually to the baseline for comparison. Finally, with the DGDH and EGDR modules added, our method achieved the best results across all metrics.

2) *Number of Neighborhood Points*: Combining adaptive propagation with normal uncertainty awareness, the depth hypothesis in weak texture areas is effectively enhanced. This

TABLE III
COMPARISON OF QUANTITATIVE RESULTS OF DIFFERENT MVS METHODS ON THE WHU-OMVS DATASET. SOME OF THE DATA IN THE TABLE ARE OBTAINED FROM AGGRMVS (BOLD FIGURES INDICATE THE BEST)

Method	MAE (m)	$\sigma < 3$ -interval (%)	$\sigma < 0.6m$ (%)
Cas-MVSNet [10]	0.1544	97.8	96.2
MS-REDNet [17]	0.1562	96.5	96.1
Ada-MVS [22]	0.1500	96.7	96.3
AggrMVS [48]	0.1534	97.1	97.0
DGE-MVSNet (Ours)	0.1085	98.0	98.8

study evaluates the impact of different numbers of neighborhood points on model performance, and the quantitative results are shown in Table V. Experiments show that setting the number of neighborhood points to 8 can achieve the best performance. In addition, to further verify the effectiveness of this method, we visualize the depth map comparison under different neighborhood point settings, as shown in Fig. 9. It can be seen that the selection of different neighborhood points will affect the depth estimation results. When the number of neighborhood points is 8, the depth map quality is the best. This is because DGE-MVSNet generates sampling points by collecting hypotheses from pixels on the same surface through adaptive propagation. Compared with uniform

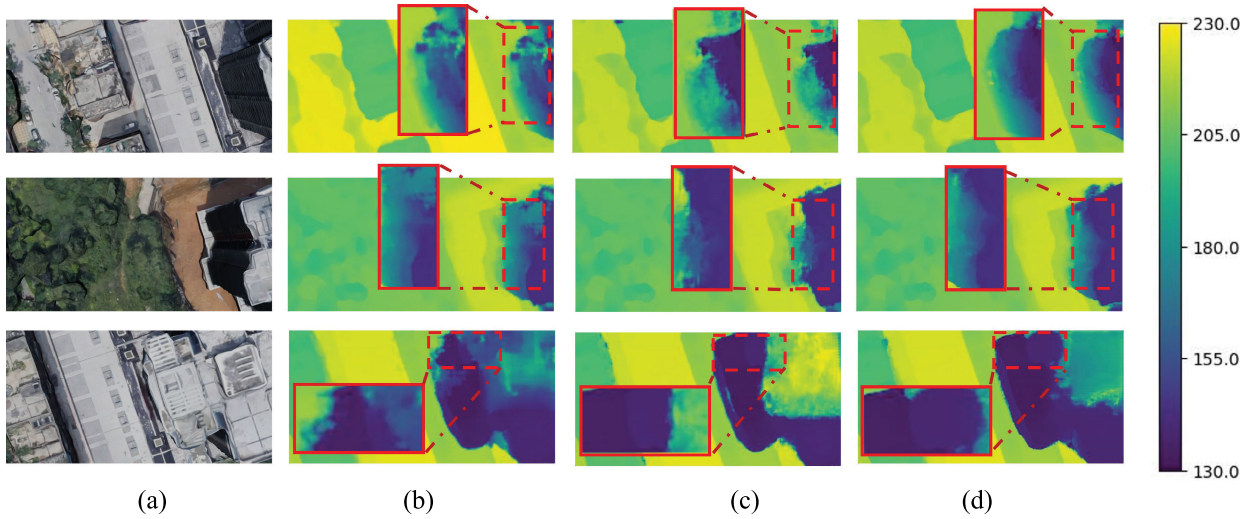


Fig. 8. Visualization on the WHU-OMVS test dataset [22]. Our method shows the best performance in the local magnified area in the red box. (a) Reference Image. (b) Cas-MVSNet. (c) Ada-MVS. (d) Ours.

TABLE IV
ACCURACY COMPARISON OF DIFFERENT MODULE SELECTIONS WITHIN THE STRUCTURE OF OUR DGE-MVSNET
(BOLD FIGURES INDICATE THE BEST)

Method	MAE (m)	$\sigma < 3$ -interval (%)	$\sigma < 0.6m$ (%)	Mem. (GB)	Run-time(s)
Baseline	0.114	96.7	97.1	1434	0.156
Baseline+ DGDH	0.083	97.5	98.6	1465	0.167
Baseline+ EGDR	0.091	97.1	97.9	1689	0.195
Baseline+ DGDH+ EGDR	0.079	98.9	99.1	1717	0.229

TABLE V
ACCURACY COMPARISON FOR DIFFERENT NUMBERS OF NEIGHBORHOOD POINTS IN DEPTH HYPOTHESES (BOLD FIGURES INDICATE THE BEST)

Number of points	MAE (m)	$\sigma < 3$ -interval (%)	$\sigma < 0.6m$ (%)
0	0.085	97.5	98.6
4	0.081	97.9	98.2
8	0.079	98.9	99.1
16	0.084	98.1	98.6

TABLE VI
ACCURACY COMPARISON FOR DIFFERENT NUMBERS OF DEPTH HYPOTHESES (BOLD FIGURES INDICATE THE BEST)

Number of depth hypotheses	MAE (m)	$\sigma < 3$ -interval (%)	$\sigma < 0.6m$ (%)	Mem. (GB)	time (s)
(40,24,8)	0.091	96.8	97.4	1321	0.168
(48,24,8)	0.082	97.3	98.4	1585	0.194
(48,32,8)	0.079	98.9	99.1	1717	0.229
(48,32,16)	0.080	98.7	98.6	2034	0.321

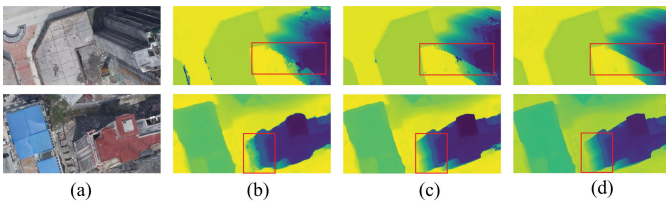


Fig. 9. Depth map visualization under different numbers of neighborhood points. (a) Reference image and (b)–(d) correspond to the depth maps generated when the number of neighborhood points is 0, 4, and 8, respectively.

sampling, this method can more effectively obtain reliable depth hypotheses. In addition, by perceiving the geometric features of depth through normal information, this strategy can generate more accurate depth estimation results compared to traditional methods that only collect depth hypotheses based on normal.

3) *Number of Depth Hypotheses*: In the process of depth hypothesis generation, appropriately increasing the number

of hypotheses can help improve the accuracy of depth estimation, but it will also bring higher video memory usage and computational overhead. To this end, we systematically explored the number of depth hypotheses in the three stages of DGE-MVSNet, aiming to achieve the best balance between accuracy and resource consumption. Table VI shows that as the number of depth hypotheses increases, both the depth estimation accuracy and memory consumption are improved. When the number of depth hypotheses is set to (48, 32, 8), it is possible to keep the memory and time overhead at a low level while ensuring high accuracy.

4) *Number of Attention Heads*: To more effectively enhance the attention to key areas (such as object edges and areas with obvious structural changes), we introduce the multihead attention to optimize the edge cost volume. Considering that the number of attention heads directly affects the representation ability of the model, we further analyze the impact of different numbers of attention heads on model performance

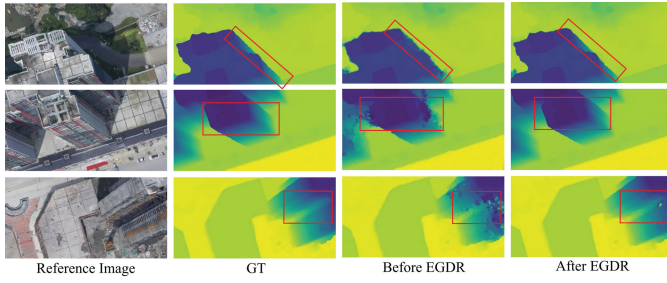


Fig. 10. Visualization of depth map before and after EGDR module.

TABLE VII

ACCURACY COMPARISON FOR DIFFERENT NUMBERS OF ATTENTION HEADS (BOLD FIGURES INDICATE THE BEST)

Number of attention heads	MAE (m)	$\sigma < 3$ -interval (%)	$\sigma < 0.6$ m (%)
0	0.085	97.8	98.8
4	0.087	98.1	98.9
8	0.079	98.9	99.1
12	0.082	98.4	98.9

in Table VII. Among them, when the number of attention heads is set to 0, it means that the attention mechanism is not used in the EGDR module, which is used as a benchmark for comparative experiments. In the experimental setting, the input channel of the CNN is $G + 1$, which is concatenated by the cost volume $C \in \mathbb{R}^{G \times D \times H \times W}$ and the coarse depth map $D_{\text{init}} \in \mathbb{R}^{1 \times H \times W}$. The output channel is set to the embedding dimension of the Transformer, which is usually 32. After being processed by the CNN, the input data is converted into a structure that meets the Transformer encoding requirements and is input into the Transformer for feature modeling. In the Transformer architecture, 0, 4, 8, and 12 attention heads are set, respectively, to evaluate the impact of different attention head configurations on model performance. Experimental results show that when the number of attention heads is set to 8, the model performance is the best, which can effectively improve the depth estimation accuracy of edge areas.

We visualized the initial depth and refined depth, as shown in Fig. 10. The comparison shows that the depth map refined by the EGDR module is closer to the GT, with more complete depth and reduced noise, which proves that the module does effectively use edge information to improve the initial depth map.

V. DISCUSSION

A. GPU and Runtime

In this section, we conduct efficiency experiments to compare the running time and memory consumption of each method. As shown in Table VIII, it can be concluded that Cas-MVSNet [10] has the shortest inference time compared with other methods. Our method outperforms the state-of-the-art method EG-MVSNet [20] in all accuracy metrics (MAE: 0.079, $\sigma < 3$ -interval: 98.9%, $\sigma < 0.6$ m: 99.1%). This is because our method effectively incorporates edge information into the initial depth map and uses the attention mechanism to capture the edge, which is very important for

TABLE VIII

COMPARISON OF ACCURACY AND EFFICIENCY OF DIFFERENT MVS METHODS ON THE WHU-MVS DATASET (BOLD FIGURES INDICATE THE BEST)

Method	MAE (m)	$\sigma < 3$ -interval (%)	$\sigma < 0.6$ m (%)	Mem. (GB)	time (s)
R-MVSNet	0.173	93.8	95.4	1133	0.216
Cas-MVSNet	0.095	97.8	97.8	1778	0.115
RED-Net	0.104	97.9	98.1	1169	0.285
EG-MVSNet	0.081	98.7	98.5	2424	0.485
DGE-MVSNet	0.079	98.9	99.1	1717	0.229

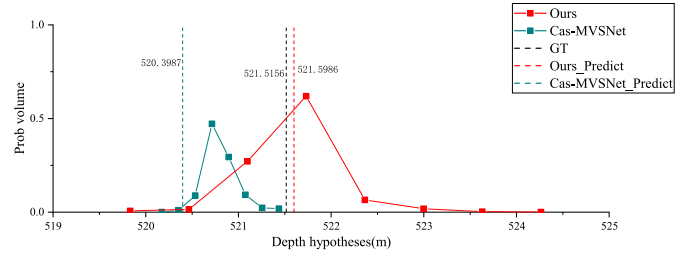


Fig. 11. Comparison of depth hypotheses between Cas-MVSNet [10] and our method in third stage. The horizontal axis is the depth hypotheses range, and the vertical axis indicates the prob volume (normalized, 0–1) of the depth hypotheses corresponding to a certain pixel. Our method produces more accurate depth hypotheses.

recovering the depth map, especially in some weak texture areas. And compared with EG-MVSNet [20], our efficiency is also significantly improved. This is mainly because our method adopts a cascade architecture and group correlation [10], [32] similar to Cas-MVSNet [11], and uses a coarse-to-fine depth prediction method, which effectively reduces the scope of depth hypothesis and greatly improves efficiency. However, compared with Cas-MVSNet [11], the time of our proposed method increases. Combined with ablation experiments, this is because our proposed DGDH and EGDR modules inevitably bring a certain time burden, including the normal depth assumption and the cosine similarity method.

B. Visualization

To verify the effectiveness of the proposed module, this section presents various visualization results, including depth hypothesis visualization, edge feature visualization, and attention weight visualization, providing an intuitive demonstration of the improvements in depth estimation.

1) *Depth Hypothesis*: A specific pixel in the reference view is selected, and its depth hypothesis range, corresponding probability distribution, and final depth estimation value are displayed. Additionally, a comparison is made with the uniform sampling method used in Cas-MVSNet [10]. As shown in the Fig. 11, the proposed method generates a more reasonable depth hypothesis, resulting in depth estimations that are closer to the ground truth. This further validates the effectiveness of the proposed DGDH strategy.

2) *Edge Feature Visualization*: To further explore the practical role of edge features in depth estimation, we conduct an intuitive visualization analysis of image features and edge features. Fig. 12 shows that edge features effectively highlight

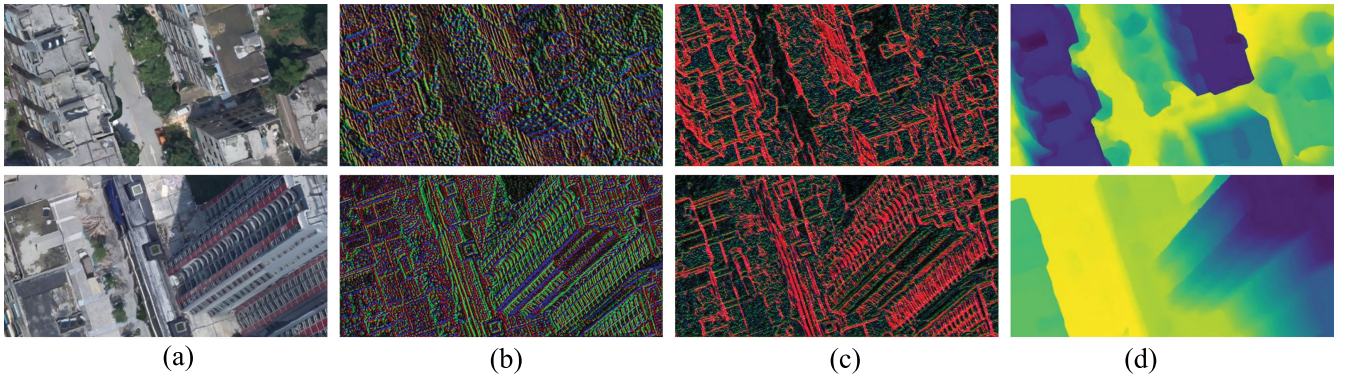


Fig. 12. Visualization of edge feature and image feature. The edge features can clearly highlight the edge information in the image and make the contour of the depth map more complete and clearer. (a) Reference Image. (b) Image feature. (c) Edge feature. (d) Depth map.

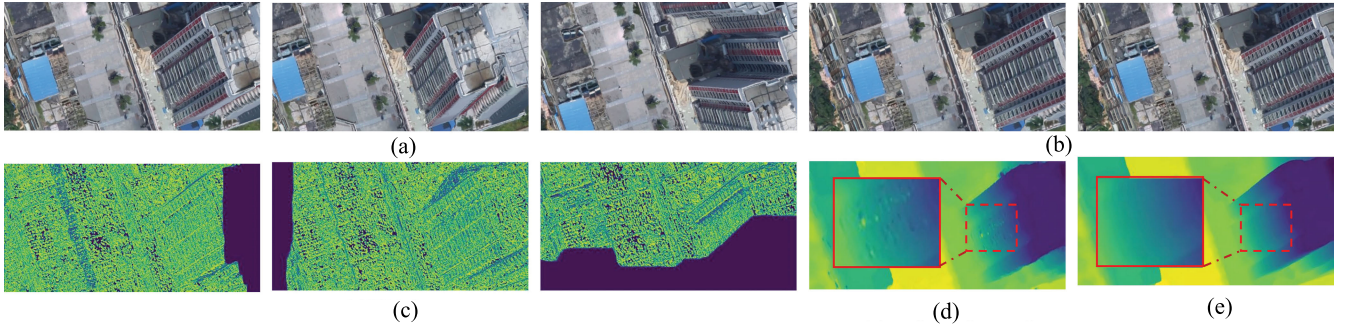


Fig. 13. Visualization of the weight map generated in the EGDR module. The three columns on the right show the source images of adjacent views with the weight map, and the column on the left shows the reference image and its corresponding depth map. (a) Source image. (b) Reference. (c) Weight map. (d) Origin depth map. (e) Filter depth map.



Fig. 14. Visualization of the point cloud and textured mesh on the WHU-OMVS testing area (a local region). (a) Reconstructed point cloud. (b) Enlarged area of the point cloud. (c) Mesh.

key structural information in the image, especially at object boundaries, regions with significant texture variations, and

occluded areas. Combined with an ablation study, it is further verified that the introduction of edge features helps to address the issue of insufficient information in weak-textured areas, which is a limitation of traditional image features. By providing geometric edge cues for the depth estimation, edge features significantly enhance the restoration of depth map details.

3) *Weight Map in EGDR*: In the EGDR module, view weights are used to filter out regions in the source views that do not match the reference view. To illustrate this concept more intuitively, we visualize the view weights, where different colors represent the varying weights assigned to each source view during the cost volume aggregation as Fig. 13. The dark areas in the weight map represent regions in the source view that cannot be matched with the reference view, and the weights are set to 0. This effectively eliminates regions that negatively affect the cost-volume aggregation, thereby improving the accuracy of the cost-volume aggregation.

C. Large-Scale Reconstruction

Our DGE-MVSNet is capable of generating full-resolution depth maps for any number of depth hypotheses planes, making it highly advantageous for large-scale multi-view aerial image reconstruction tasks. Running on a 24GB GPU, the model performs depth map prediction on the WHU-OMVS testing set, where each image has a resolution of 3712×5504 pixels. It takes only 3.8 s to infer 384 depth hypotheses planes, and generate the depth map. As shown in Fig. 14, the reconstruction results cover an area of approximately 0.85×0.70 km². The entire scene reconstruction takes about

48 min, compared to 151 min for COLMAP [7], highlighting the efficiency and practicality of DGE-MVSNet in real-world applications.

VI. CONCLUSION

In this study, DGE-MVSNet is proposed, which includes a DGDHs model designed to provide reliable depth hypotheses, and an EGDR module designed for edge information of large-scale aerial MVS. The DGDH module builds a set of neighborhoods for points in the reference view, ensuring that any point in the neighborhood lies on the same physical surface. Relying on this neighborhood set, the initial normal depth hypotheses can be estimated from these points. The uncertainty awareness of these neighborhood points is then weighted and combined with the uniform hypotheses to dynamically adjust the normal depth hypotheses. To address the challenge of matching in weak texture regions, the EGDR module utilizes edge feature volumes to guide cost volume aggregation. It calculates view weights and generates invalidation masks to effectively filter out mismatched regions between source and reference views. During depth refinement, the depth estimation in the coarse stage is enhanced by adaptively fusing the edge cost volume with the initial depth and leveraging an attention mechanism.

Evaluation results on the WHU-MVS dataset show that our method reduces the MAE by 50.1% compared to MVSNet, with improvements of 24.0% over similar methods, RED-Net, while also significantly improving efficiency. To further demonstrate the generalization of our method, we also evaluated it on the Luojia-MVS dataset and the WHU-OMVS dataset. The experimental results on Luojia-MVS and WHU-OMVS show that our method reduces MAE by 4.3% and 29.2%, respectively, compared with the state-of-the-art methods EG-MVSNet and AggrMVS. Comprehensive ablation experiments further confirm the effectiveness of the DGDH and EGDR modules for large-scale aerial MVS applications. However, our approach currently depends on a predefined method for extracting edge features. In future work, we plan to investigate more learnable techniques to enhance its adaptability. Furthermore, a deeper analysis of the key MVS parameter view overlap is essential to ensure high-quality 3-D reconstruction, particularly in environments with limited multi-view overlap or unstructured conditions.

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